



Phylogeny of *Acomys spinosissimus* (Rodentia, Muridae) from north Malawi and Tanzania: evidence from morphological and molecular analysis

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New specimens of *Acomys spinosissimus* from northern Malawi and southern Tanzania were subjected to morphological, molecular (cytochrome *b* mitochondrial gene) and cytogenetical analysis in order to establish the phylogenetic position of these populations among *A. spinosissimus*, as well as their relationships with other species (*A. subspinosus* and *A. wilsoni*). The monophyly of the *spinosissimus* clade is supported by morphological and molecular phylogenies. Cytogenetics and morphology, but not the cytochrome *b* phylogeny show the possible existence of two sibling species within *spinosissimus*. Incongruences are noted between molecular and morphological trees, especially concerning the position of the Transvaal specimens. Two different biogeographic scenarios for the origin and dispersal of *A. spinosissimus* are discussed in the light of available palaeontological data.

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ADDITIONAL KEY WORDS: *Acomys* – systematics – morphology – cytochrome *b* – cytogenetics – phylogeography – Africa.

INTRODUCTION

The spiny mouse (*Acomys* Geoffroy Saint-Hilaire, 1838) has recently been subject to molecular and morphological analyses due to its controversial phylogenetic position among the Muroidea (Sarich, 1985; Denys *et al.*, 1992; Chevret *et al.*, 1993; Chevret & Hänni, 1994; Hänni *et al.*, 1995). Present in Africa, the Middle East, Arabia and Pakistan, the genus has a peculiar biogeographic distribution compared to other African true Murinae. Despite recent taxonomic revisions by Musser & Carleton (1993) and Denys *et al.* (1994), its systematics is still in need of revision.

A. spinosissimus Peters 1852 has a distribution ranging from southern Tanzania to southern Transvaal, which corresponds to the Zambezi domain. Southern Tanzania and northern Malawi are the boundary between rodent species from the northern (Somali-Masai

& Sudano-Guinean) and the southern savannahs (Zambezian). Northern Malawi is also the boundary between northern and southern species (Ansell & Dowsett, 1988). Other species are widespread in Malawi; *Acomys spinosissimus* is such a species and it is important to verify the status of populations in that region as well as their phylogenetic affinities. Moreover, South African *Acomys* fossils from as early as 5 Myr have been related to this group (Denys, 1990). We need a better understanding of the evolution of this species and the role of the northern Malawi region as a contact zone.

The cytochrome *b* mitochondrial gene was chosen as a molecular marker because it is commonly used to investigate phylogenetic relationships between rodent species (Lessa & Cook, 1998; Matthee & Robinson, 1997; Myers, Lindrigan & Tucker, 1995; Smith, 1998; Smith & Patton, 1993; Wettstein *et al.*, 1995) and has been informative about interspecific relationships within *Acomys*. It has confirmed hypotheses based on

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cytogenetical analysis about the plesiomorphic position of *A. spinosissimus* among other species of *Acomys* (Barome, Monnerot & Gautun, 1998).

The present study focused on recently collected specimens of *A. spinosissimus* from northern Malawi and southern Tanzania. We used morphological, molecular and cytogenetical analyses in an attempt to assess the phylogenetic relationships between the populations and other *spinosissimus sensu lato* populations. We discuss the monophyly of *A. spinosissimus*, estimate the dates for the differentiation of the lineages and propose two phylogeographic hypotheses.

MATERIAL AND METHODS

SPECIMENS UNDER STUDY

Besides *A. spinosissimus*, specimens of *A. subspinosus* Waterhouse 1838 and *A. wilsoni* Thomas 1892 were examined. Two murinae (*Aethomys*, *Mus*), another Acomyinae (*Uranomys*), a Gerbillinae (*Tatera*) and a Dendromurinae (*Steatomys*) were used as outgroup taxa following previous morphological (Denys *et al.*, 1992, 1994, 1995) and molecular (Chevret *et al.*, 1993; Chevret & Hänni, 1995) results. All information about specimens is summarized in Table 1, and geographical positions of localities are shown in Figure 1. Some specimens (see Table 1) were examined because their karyotypes were known: one specimen from Ifikara and the series from Blantyre, in Matthey (1965), identified by Petter as *A. selousi* De Winton 1896, the series from Shamdungila and the specimens from Levuvhu and Pafuri from Dippenaar & Rautenbach (1986).

MORPHOLOGICAL ANALYSIS

Classical morphometrics on selected skull characters were conducted following the method described by Denys & Tranier (1992). Specimens showing extreme dental wear (stages 1 and 6) were not included. The dental nomenclature and abbreviations follow Denys *et al.* (1992). Phylogenetic treatment of morphological characters were performed under PAUP 3.1 (Swofford, 1993) under a heuristic search option.

DNA SEQUENCING AND PHYLOGENETIC RECONSTRUCTIONS

DNA extraction, amplification of the cytochrome *b* mitochondrial gene (cyt *b*) and its sequencing were performed as described by Barome *et al.* (1998). The cyt *b* gene was sequenced for nine *Acomys* (see Table 1 for details) from four different species: *A. subspinosus*, *A. spinosissimus*, *A. wilsoni*, and *A. dimidiatus* Cretzschmar 1826. We also used previously published sequences (Barome *et al.*, 1998, #1, 2, 7, 9 and #14 to 19) from *A. spinosissimus*, *A. russatus*

Wagner 1840, *A. ignitus* Dollman 1910, *A. cahirinus* Desmarest 1819, *A. airensis* Thomas et Hinton 1921 and *Acomys* sp. The outgroup is composed of cyt *b* sequences of *Mus musculus* Linnaeus 1758 (Bibb *et al.*, 1981), *Tatera kemp* Wroughton 1906 collected in Lamto (Ivory Coast, 06°12'N, 04°58'W), voucher and *Steatomys* sp. Peters 1846 collected in Bankoumana (Mali, near Bamako, 12°12'N, 08°16'W), voucher number MNHN-CG 1998-1226).

Three different methods were used for phylogenetic reconstruction: Maximum Parsimony, Neighbor-joining (Saitou & Nei, 1987) and Maximum Likelihood (Edwards, 1972). Corresponding programs used were PAUP 3.1.1 (Swofford, 1993) and RNA (Farris, 1994), MEGA 1.01 (Kumar, Tamura & Nei, 1993), and Puzzle 3.1 (Strimmer & von Haeseler, 1997). We use the following search procedures: PAUP: branch & bound search bootstrap with 100 replicates, ignoring invariant characters; RNA: 1000 bootstrap replicates; Puzzle: 1000 puzzling steps, Tamura-Nei model (Tamura & Nei, 1993), mixed heterogeneity rate (invariant + gamma distributed rate); MEGA: 100 bootstrap replicates, neighbor-joining, distance used: Kimura-2 parameter (Kimura, 1980).

CYTOGENETICS

A male specimen from Berega was analysed. Chromosome preparations were made from fibroblast culture established after skin biopsy. The chromosomes were studied by G-banding (GTG; Seabright, 1971), by R-banding or BrdU incorporation (RBG: Viégas-Péquignot & Dutrillaux, 1978) and by C-banding (CBG; Sumner, 1972; ISCN, 1995). At least ten metaphases were analysed by each of the techniques. Cryopreserved fibroblast cultures and tissue explants of the specimen examined in this study are available in the cryopreserved cell and tissue bank of the MNHN.

RESULTS

MORPHOLOGICAL STUDY

Comparison between new specimens (Berega and Mwenerondo) and others from Malawi and Tanzania

The newly collected specimens from Mwenerondo (north Malawi; MNHN-CG 1999-74) and Berega (Tanzania; MNHN-CG 1998-1611) are nearly the same size and apparently very similar (Fig. 2). However, there are some slight differences compared to the Malawian specimen: the skull from Tanzania has a more robust braincase, shorter rostrum and incisor foramen, wider inter t4 region, more parallel teeth rows, smaller and less oblique tympanic bullae (Appendix 1). The external characteristics are very similar except that the specimens from Tanzania have a shorter tail, shorter hind

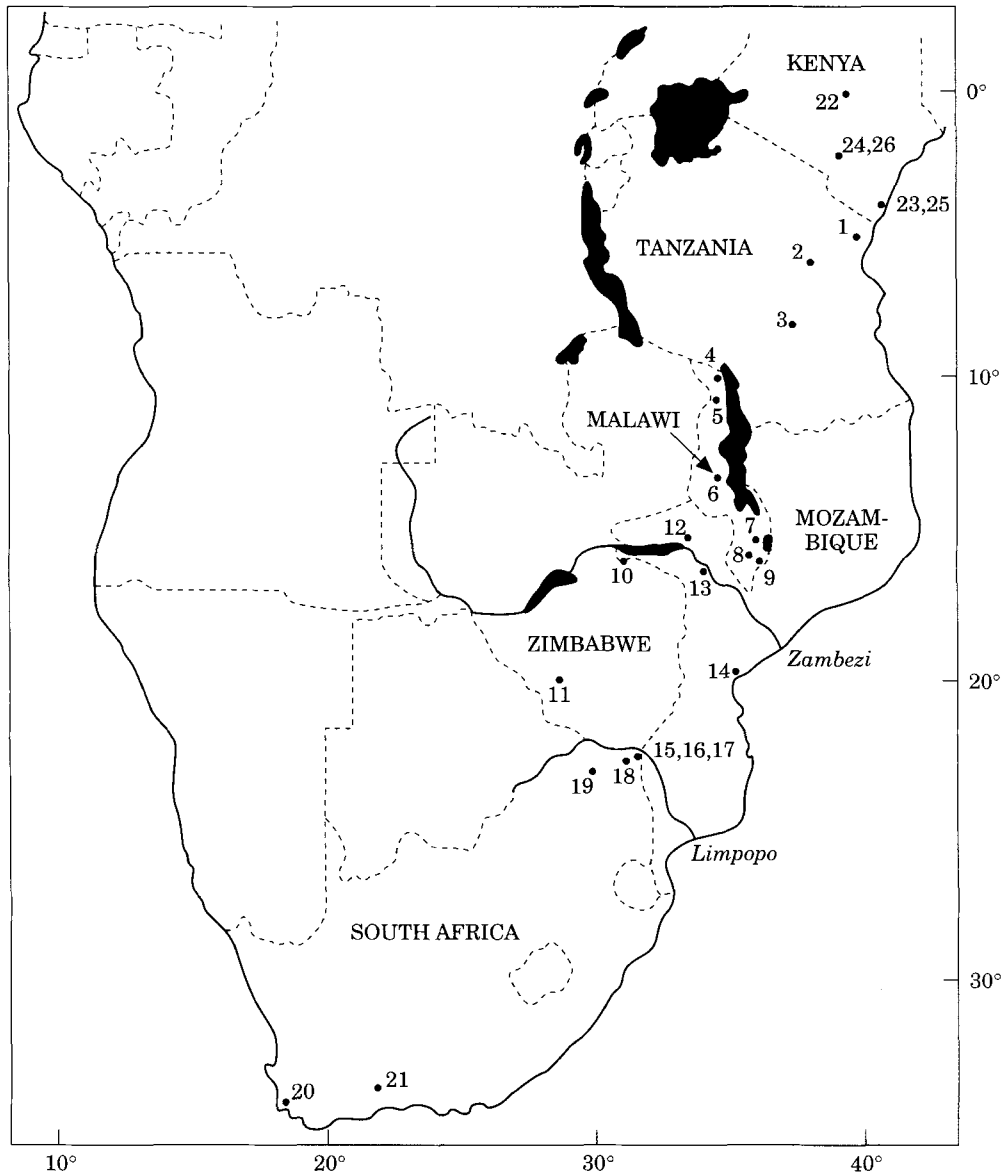


Figure 1. Geographical provenance of the specimens examined in this study. Locality numbers refer to Table 1.

feet, a darker coloration of the dorsal pelage and more reddish flanks. The size of the molars is the same in both specimens despite slight differences in morphology. The Tanzanian specimens have a less bunodont aspect of the cusps and a narrower upper first molar. The new Malawian specimen is more worn than the Tanzanian one and has a trace of posterior cingulum on the upper M/12, a second cusp on the posterior lobe of the lower M/3 and an anterolabial cingulum on M/2 (Fig. 3).

The new specimens are similar in morphology and skull measurements to those from Tanzania and Malawi. They have a large Cp on the upper M/1 as well as a well developed cv1–cv5 on the lower molars. This cusp disposition is similar to that displayed by

karyotyped specimens from Blantyre. The specimens from Liwonde (south Malawi) also show an elongated t1 and well individualized cusps of the prelobe of the lower M/1, which is not the case for the karyotyped specimen of Matthey (1965) from Ifikara, Tanzania (Fig. 3).

Comparison with A. spinosissimus type and type series

The new specimens were compared with the lectotype of *A. spinosissimus* from Buio (Mozambique) and with karyotyped specimens from Dippenaar & Rautenbach (1986). According to these authors, “the lectotype [is] in the upper range of size variation of *spinosissimus*”

Table 1. List of the *Acomys* material used in the present work including locality coordinates and specimen numbers

Species	Country	Locality	No ¹	Coordinates	Voucher number ²	Mor ³	Cyto ⁴	Prot ⁵	Cyt b ⁶	Voucher label
<i>A. spinosissimus</i> Peters 1852 <i>sensu lato</i>	Tanzania	Amani (5 sp)	1	05°09'S–38°36'E	NHM serie 761 1559 to 1563	Yes	—	—	—	<i>A. spinosissimus</i>
		Berega (2 sp)	2	06°10'S–37°12'E	MNHN-CG 1998-1611	Yes	Yes	—	AJ010558 AJ010559	undiagnosed undiagnosed
		Ifikara (1 sp)	3	08°10'S–36°38'E	RUCA Tanz. coll 14621	Yes	—	—	—	cf. <i>A. selousi</i> ^b
	Malawi	South, unknown locality		unknown	MNHN-CG 1964-53	Yes	M	—	—	unknown
		Mwenzerondo	4	09°54'S–33°55'E	NHM 84.1842	Yes	—	—	—	undiagnosed
		Nyika Plateau (3 sp)	5	≈ 10–11°S–34°E	MNHN-CG 1999-74 Blantyre, Coll. 96-36 NHM 97.10.1	No	—	—	AJ010556 idem	undiagnosed undiagnosed
		Ntchisi Mt	6	13°22'S–34°01'E	208, 285 to 287 NHM 49.378	Yes	—	—	—	<i>A. selousi</i>
		Liwonde National Park (3 sp)	7	15°03'S–35°15'E	TM 41760-62	Yes	—	—	Z96067*	?
		Blantyre (9 sp)	8	15°46'S–35°00'E	MNHN-CG 1964-303, 883 to 891	Yes	M	—	—	cf. <i>A. spinosissimus</i> cf. <i>A. selousi</i> ^b
		Mulanje, Luncheon river (2 sp)	9	16°02'S–35°30'E	NHM 14.6.13.31 & 32	Yes	—	—	—	<i>A. selousi</i>
	Zimbabwe	Angwa Bridge	10	15°51'S–30°35'E	MNHN-CG 1990-592	Yes	—	—	—	<i>A. spinosissimus</i>
		Kudzwe Dam		unknown	MNHN-CG 1990-352	Yes	—	—	—	<i>A. spinosissimus</i>
		Bulawayo, Essex farm (5 sp)	11	20°10'S–28°43'E	NHM 97.1.4 44 to 48 (type series)	Yes	—	—	—	Type <i>A. selousi</i> (DeWinton 1897)
	Mozambique	Songo (2 sp)	12	15°36'S–32°45'E	ISEK M/10141/90, M/10127/90	Yes	—	—	—	<i>A. spinosissimus</i>
		Buio	13	16°10'S–33°35'E	ZMB 1711	Yes	—	—	—	Lectotype <i>A. spino</i>
	South Africa	Beira (10 sp)	14	19°49'S–34°52'E	NHM 7.6.2.80 to 89	Yes	—	—	—	<i>A. selousi</i>
		Pafuri, Kruger National Park	15	22°25'S–31°07'E	TM 29879	Yes	D&R	—	—	<i>A. spinosissimus</i>
		Pafuri, Kruger National Park	16	22°27'S–31°21'E	TK 55186 (Dna 33626)	—	—	J (no 1)	Z96068**	<i>A. spinosissimus</i>
		Levuvhu hippo pool (Siunuwana ranger post)	17	22°26'S–31°11'E	TM 30520	Yes	D&R	—	—	<i>A. spinosissimus</i>
		Shamdungila, Kruger National Park (6 sp)	18	22°38'S–30°56'E	TM 30577 to 579, TM 30614 to 616	Yes	D&R	—	—	<i>A. spinosissimus</i>
		Newgate	19	22°58'S–29°56'E	TM 3593	Yes	—	—	—	Type <i>transvaalensis</i> (Roberts 1926)

continued

Table 1 – continued

Species	Country	Locality	No ⁱ	Coordinates	Voucher number ^b	Mor ^c	Cyto ^d	Prot ^e	Cyt b ^f	Voucher label
<i>A. subspinosus</i> Waterhouse 1838	South Africa	Table Mounts, Cape Swartberg Pass	20 21	33°58'S–18°25'E 33°31'S–22°03'E	NHM 55.12.24.147 TM 40893	Yes	—	— J (no 2)	— AJ010557	Type <i>subspinosus</i> <i>A. subspinosus</i>
<i>A. wilsoni</i> Thomas 1892	Kenya	Rojewero Plains, Meru National Park (6 sp)	22	00°11'N–38°10'E	MNHN-CG 1982-204 to 209	Yes	—	—	—	<i>A. wilsoni</i>
		Shimba Hills National Reserve	23	04°15'S–39°23'E	CM 98234 (Dna 33216)	—	—	J (no 20)	AJ010560	<i>A. wilsoni</i>
		Bushwhackers, Machakos district	24	02°19'S–38°07'E	CM 98239 (Dna 33371)	—	—	J (no 21)	AJ010561	<i>A. wilsoni</i>
<i>A. ignitus</i> Dollman 1910	Kenya	Moana Marine Station, Kwale district	25	04°18'S–39°35'E	TK 98216 (Dna 33298)	—	—	J (no 5)	Z96063*	<i>A. ignitus</i>
		Bushwhackers, Machakos district	26	02°19'S–38°07'E	TK 98223 (Dna 33367)	—	—	J (no 7)	Z96064*	<i>A. ignitus</i>
<i>A. russatus</i> Wagner 1840	Jordan Israel	Lava Hills, S of Azraz ed Druz En Gedi, Dead Sea	27 28	unknown 31°27'N–35°23'E	TK 40628 (Dna 25598) JCG-Orstom 95-011	— —	Q —	J (no 11) —	Z96066* Z96065*	<i>A. russatus</i> <i>A. russatus</i>
<i>A. dimidiatus</i> Cretzschmar 1826	Jordan Israel	Aqaba Bethlehem, occupied West Bank	29 30	29°31'N–31°00'E 31°42'N–35°13'E	TK 40619 (Dna 25618) TK 40646 (Dna 25636)	— —	Q —	J (no 12) J (no 14)	AJ010554 AJ010555	<i>A. cahirinus</i> <i>A. cahirinus</i>
<i>Acomys</i> sp.	Burkina Faso Cameroon	Banfora Garoua	31 32	10°38'N–04°46'W 09°17'N–13°22'E	F. Catzeflis T1058 JCG-Orstom 97-235	— —	— —	— —	Z96056* Z96058*	<i>Acomys</i> sp. <i>Acomys</i> sp.
<i>A. cahirinus</i> Desmarest 1819	Egypt	Cairo	33	30°03'N–31°15'E	JCG-Orstom 94-571	Yes	—	—	Z96051*	<i>A. cahirinus</i>
<i>A. aircensis</i> 1921 Thomas & Hinton	Niger	Hamdara	34	13°49'N–09°47'E	JCG-Orstom 95-015	Yes	—	—	X96996*	<i>A. aircensis</i>

* Abbreviations of museums and collections; NHM: Natural History Museum (London); MNHN: Muséum National d'Histoire Naturelle (Paris); RUCA: University of Antwerp, Tanzania collection; TM: Transvaal Museum (Pretoria); ISEK: Institute of Evolution and Systematics (Krakow); ZMB: Zoologisches Museum Berlin; TK: Texas Tech University (Lubbock); CM: Carnegie Museum (Pittsburgh); F: Catzeflis; collection François Catzeflis, University of Montpellier; JCG-Orstom: collection Jean-Claude Gautun-ORSTOM, Centre of Niamey.

^b Specimens examined in the morphology analysis.

^c Specimens where the karyotype is known; Yes: present work; M: from Matthi (1965); D+R: from Dippenaar & Rautenbach (1986); Q: from Qumsiyeh *et al.* (1986).

^d Specimens studied with protein polymorphism by Janacek *et al.* (1991), locality number in brackets.

^e Specimens from which cyt b has been sequenced; the number is the GenBank accession number; sequences marked * are from Barome *et al.* (1998).

^f Previously mislocated in Barome *et al.* (1998) as coming from 'Nyala reserve, South Africa'.

^g Specimens diagnosed by F. Petter.

^h Localities 27 to 34 are not plotted on Fig. 1.

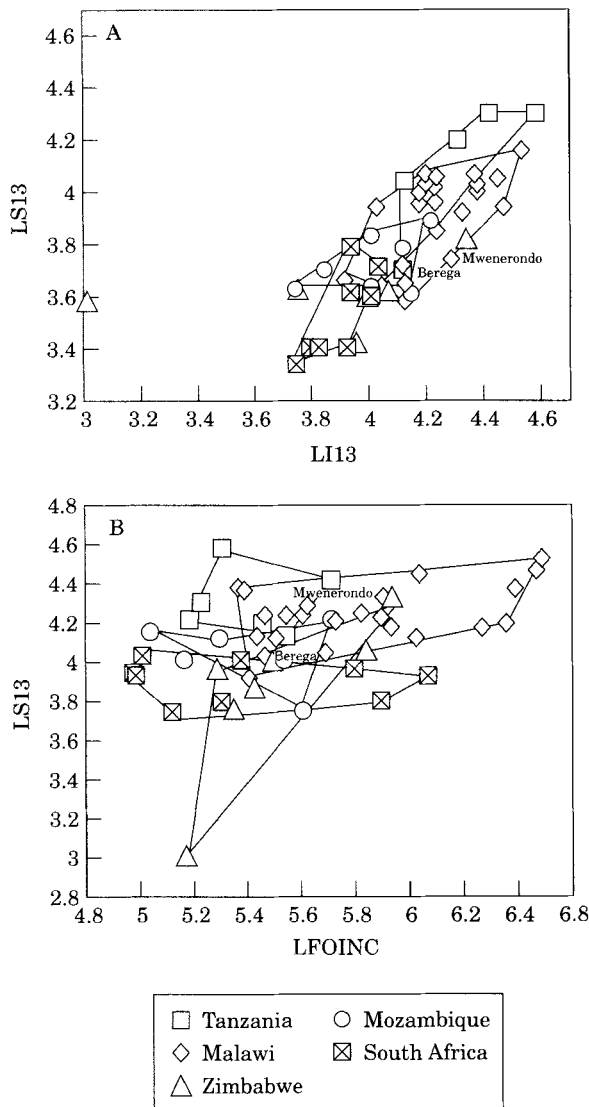


Figure 2. Bivariate plots of skull measurements in the different *A. spinosissimus*. A, LI13, lower molar row length versus LS13 (upper molar row length) shows that the specimens from Malawi and Tanzania are larger on average than those from the Transvaal, Zimbabwe and Mozambique. B, same result is obtained with a longer incisor foramen (LFOINC) for specimens from Malawi and lengthier teeth rows for those from Malawi and Tanzania compared to others. Note the proximity between the two new specimens from Berega (1998-1611) and Mwenerondo (MNHN-CG 1999-74).

specimens that were used in their study. These authors underlined that this observation, added to “uncertainty as to the position of the type locality causes some unease in our continued use of *spinosissimus* as the senior synonym”, and we agree with this remark. Moreover, Dippenaar & Rautenbach (1986) mentioned the poor state of preservation of the Buio skull (housed

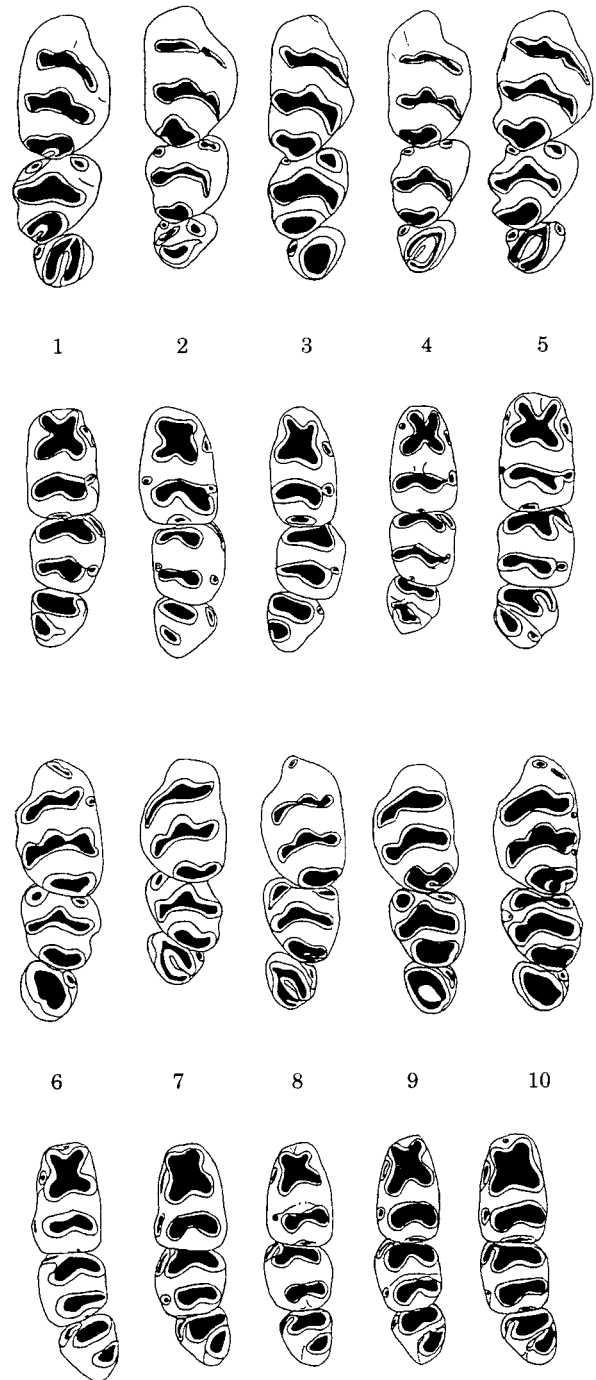


Figure 3. Molar morphology of *A. spinosissimus* from the different southern regions described here. The upper molars are rows 1 and 3 and the lower molars on rows 2 and 4. (1) 96-37 Mwenerondo North Malawi; (2) 1964-53: Ifikara, Tanzania; (3) 1998-1611, Berega, South Tanzania; (4) TM-41762, Liwonde (South Malawi); (5) 1964-886, Blantyre, S. Malawi; (6) *A. selousi*, Matabiti, BMNH-97.1.4.46; (7) *A. spinosissimus* Beira, BMNH-7.6.2.82; (8) TM-29789 Pafuri, Transvaal; (9) TM 30615 Transvaal; (10) TM30579 Shamdigula, Transvaal.

today in the Zoologisches Museum Berlin), and the difficulty in taking accurate measurements from it.

We observed that the hind foot of the type is very large, and the pelage colour very dark with a grey underpart. The dental rows show a very worn stage (stage 5) and the disposition of the cusps is difficult to discern.

The specimens from Songo (close to Tete, the locality of the missing type), all have dental morphology with a narrow elongated longitudinally oriented t1. Songo specimens show a posterior crest extending the cusps backwards, and the posterior cingulum on the upper M/1 is not seen. On the lower M/1 there is no Cp on the M/2, and the cv5 is not as well developed as the antero-labial cingular crest. The new specimens from Tanzania and north Malawi display a t1 and t4 with a posterior crest and are longitudinally oriented as in the Mozambique Songo specimens (Fig. 3).

Comparison with A. selousi type series

When compared to the *A. selousi* type series from Bulawayo (Zimbabwe), the specimens from Malawi and Tanzania have the same grey to white tail and a grey dorsal pelage. The *selousi* type series specimens have smaller HB, HF, WNAS, LS13/, and LI13, compared to those of Malawi and Tanzania (Appendix 1, Fig. 2). The type specimen from Matabiti (Essex farm, near Bulawayo, Zimbabwe) is nearly complete (NHM) and reveals advanced wear of the molars (stage 4). According to Dippenaar & Rautenbach (1986), this type "falls well within the range of variation of the Transvaal [specimens]". From our observations, the type of *selousi* is outside of the range of the specimens from Malawi and Tanzania and the type of *A. transvaalensis* Roberts 1826 is very small (Appendix 1, Fig. 2). Specimens from the type locality of *selousi* display bunodont t1, and t4 is not very elongated with no posterior longitudinal crest. These specimens are characterized by the development of cingular cusps on the prelobe of the M/1 and the posterior cingulum is not visible. On the lower molars, the cv5 is small but there is a tma and a cv1. The anterolabial crest is short (Fig. 3).

Comparison with A. transvaalensis holotype

The holotype of *A. transvaalensis* from Newgate shows a very worn dental pattern (stage 5). According to Dippenaar & Rautenbach (1986), it "falls well within the range of colour variation of *Acomys* from the Transvaal and Zimbabwe", and they put it in synonymy with *A. spinosissimus* despite the rather poor state of preservation of the type of this species. The North Malawian specimen has a yellow pale brown dorsal pelage, while the South African specimens have a more reddish one, with strongly orange flanks. It also has

a white tail identical to the Tanzanian, the south Malawian and *selousi* specimens, while the other South African specimens have a yellow tail.

The other Transvaal specimens studied here differ in the general size of the skulls. The old specimens from Transvaal (stage of wear 5) have the same size as younger (stage 3) North Malawian and Tanzanian specimens. Figure 2 and data in Appendix 1 also confirm the smaller skull size of the Transvaal specimens compared to those from Malawi and Tanzania. In ventral view, the tympanic bullae are slightly smaller in the Transvaal specimens but there is no difference either in the length and extension of the incisor foramen or in the palate structure. The Shamdigula specimens exhibit a very bunodont pattern of the cusps molars cusps with a posterior cingulum well differentiated on upper M/1 and very bunodont t4; on the lower M/1, the two anterior cusps of the prelobe are well fused together (Fig. 3).

Comparison with the Mozambique series

The position of the Mozambique specimens seems more intermediate. Those from Songo and Buio have the same characteristics as the new specimens from Tanzania and Malawi, but the Beira specimens show a distinct molar cusp pattern (t4 without a well marked posterior crest). Two distinct taxa may therefore be present in Mozambique, but the sample is too small to be certain.

Conclusion of the morphological study

The new specimens from Malawi and Tanzania have similar characteristics which suggest they are in the same group. They share with the *A. selousi* type series the same pattern of coloration of the tail (which is white beneath), and a relatively large size of the skull. The Transvaal specimens are different from those from Malawi and Tanzania in their smaller size, HF length, molar row dimensions, cusp morphology and colour of the tail. They have a more bunodont pattern of the dental cusps which is reminiscent of *A. subspinosus*.

The observed differences between the northern (Malawi-Tanzania) and southern (Transvaal) *A. spinosissimus* could either be explained by intra-specific clinal variability with an increase of size from south to north, or by inter-specific differences. The poor state of preservation of the type specimens does not help to solve the problem.

PHYLOGENETIC ANALYSIS ON MORPHOLOGICAL CHARACTERS

The morphological study has allowed us to separate representatives of the *A. spinosissimus* group on the basis of skull and external characteristics. In order to

analyse the phylogenetic relationships among these *A. spinosissimus sensu lato* and to compare morphological and molecular results, a phylogenetic analysis was performed on external, skull and dental characters for all the specimens listed in Table 1. A character matrix has been established from the results of the morphological analysis, together with characters described by Hutterer, Lopez-Martinez & Michaux (1988) and Denys *et al.* (1992, 1995). *A. subspinosus* from Cape province and *A. wilsoni* from East Africa have been chosen because of earlier work by Janecek, Schlitter & Rautenbach (1991) and Denys *et al.* (1994) (Fig. 4A, B). For quantitative characters, we used average measurement for each population (Appendix 1). To be consistent with the molecular analysis and previous work on *Acomys*, we have used as outgroups true Murinae (*Aethomys*, *Mus*), *Acomyinae* (*Uranomys*, see Hänni *et al.*, 1995), *Dendromurinae* (*Steatomys*) and *Gerbillinae* (*Tatera*).

Populations were labelled by their geographical location rather than their species or subspecies name, in order to investigate the phylogeographic relationships between all the *spinosissimus sensu lato* without any *a priori* grouping. The north and south Malawi specimens were grouped together. The Zimbabwe collection contains the *A. selousi* type series.

Characters and their states (see character matrix Appendix 2)

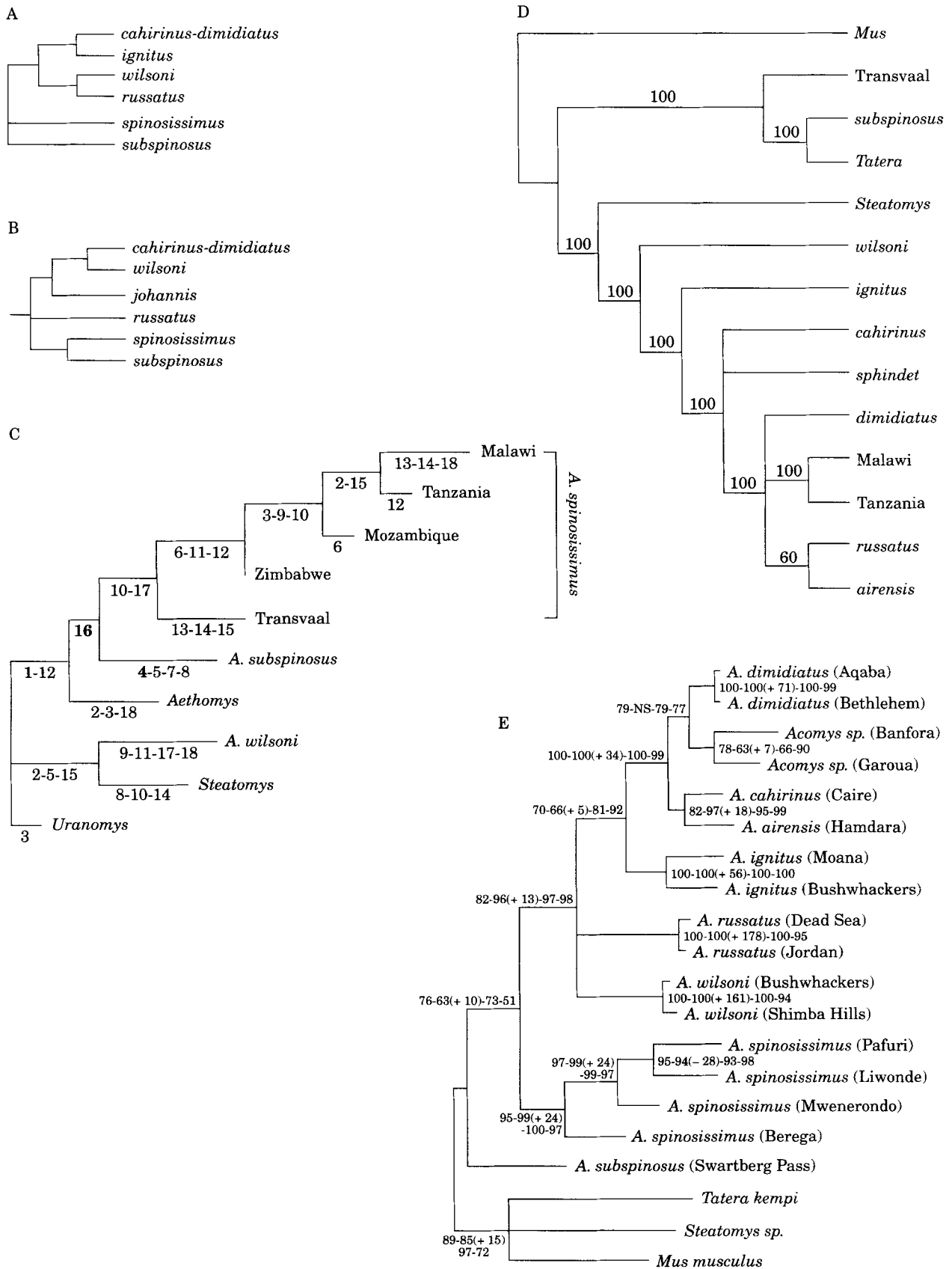
- (1) Size of the tail compared to head and body length: TL < HB (0), TL = HB (1).
- (2) Skull size lower than 24 mm (0), between 24 and 26 mm (1), greater than 26 mm (2).
- (3) Molars size (average value) lower than 4.0 mm (0), greater (1).
- (4) Incisor foramen stop before prelobe of M1/ (0), at the level of t1 or t4 (1).
- (5) Jugal arch narrow (0), wide (1).
- (6) Tail coloured (0), white (1).

- (7) Normal proportions of molars (0), microdontia (ratio of LS13/LGT < 0.15) (1).
- (8) T1 and t4 absent (0), t1 and t4 low bunodonts not well fused to t2 and t5 (1), higher fused (2).
- (9) T1 and t4 absent (0), t1 and t4 present: without crests or cusps (1) or with posterior crests and some accessory cusps (2).
- (10) Accessory cusps on M/1 prelobe absent (0), variable or present (1).
- (11) T6 smaller than t5 (0), as big or larger (1).
- (12) Cp on M/1 present (0), absent (1).
- (13) Cp on M/2 small (0), absent (1).
- (14) Cp on M/1 small (0), absent (1).
- (15) Cp on M/2 small (0), absent (1).
- (16) Connection of the anterior lobe of M/1 strong (0), weak (1), absent (2).
- (17) Anterolabial cusps of M/2 crestiform (0), present (1), absent (2).
- (18) Anterior cusps of the prelobe fused or not individualized (0), well individualized (1).

Analysis of the morphological trees

From the first set of morphological data (10 taxa, 18 characters), including all specimens of *A. spinosissimus*, and only *A. subspinosus* and *A. wilsoni* as representatives of other species of the genus, a single most parsimonious tree (Fig. 4C) was obtained (length: 39, CI: 0.487, CI excluding uninformative characters: 0.459; RI: 0.512; RC: 0.250). There were 31 trees with a length of one additional step (length: 40), which reflects the high homoplasy level of the dataset. In the shorter tree, *A. subspinosus* and *A. spinosissimus* are sister groups. Within *A. spinosissimus*, the clade Zimbabwe–Mozambique–Tanzania–Malawi is supported by the colour of the tail (character #6, with a reversion for Mozambique specimen). Sister grouping of specimens from Tanzania and Malawi is supported by the larger size of the skull (character #2), a state which is also present in *Aethomys*. The *A. spinosissimus* clade

Figure 4. (opposite) Phylogenetic relationships of *Acomys spinosissimus* and *A. subspinosus* from previous and present studies. A, phenogram based on protein polymorphism from Janecek *et al.* (1991); B, tree deduced from a scenario proposed by Denys *et al.* (1994); C, cladogram based on the present morphological study (10 taxa, 18 characters), concerning *A. spinosissimus sensu lato*, *A. subspinosus* and *A. wilsoni*; *Aethomys*, *Steatomys* and *Uranomys* were used as outgroup. The position of *A. wilsoni* is artefactual because of homoplasy. The tree has a length of 39, a CI of 0.487 (0.459 when uninformative characters are excluded), a RI of 0.512 and a RC of 0.250. Character changes are indicated above each branch; the characters that are not homoplastic are typed in boldface. D, majority-rule consensus tree from the 10 most parsimonious trees obtained from morphological data, using the same specimens as cytochrome *b* phylogeny (14 taxa, 18 characters) (treelength: 54, CI: 0.426, RI: 0.551, RC: 0.235). E, majority-rule bootstrap consensus phylogram from cytochrome *b* sequences (1141 bp). Support values obtained with PAUP (maximum parsimony, branch & bound search, 100 bootstrap replicates), RNA (maximum parsimony, 1000 bootstrap), MEGA (Neighbor-joining, Kimura-2 parameter distances, 100 bootstrap) and PUZZLE (maximum likelihood, 1000 puzzling steps) are shown near each branch in this order (for RNA, bootstrap and support index are separated by a slash). Tree length is 908 (PAUP, b&b search, length of the next shortest trees: 918), CI with uninformative characters is 0.688, without them 0.601.



is supported by two synapomorphies, which are also present in *Steatomys* (character #10) and *A. wilsoni* (character #17). The clade grouping *A. subspinosus* and *A. spinosissimus sensu lato* is supported by one synapomorphy, the weak connection of the anterior lobe of the M/1 (character #16). Notice here the artefact position of *A. wilsoni*, which clusters in the outgroup with *Steatomys* because of the high amount of homoplasy.

The second morphological dataset is based on the same specimens as the molecular dataset and thus can be directly compared with it. The analysis based on 14 taxa and 18 characters (Appendix 2) yields ten most parsimonious trees (length: 54, CI: 0.426, RI: 0.551, RC: 0.235), the majority rule consensus tree of which being presented in Figure 4D. In this tree the *A. spinosissimus* specimens do not cluster in a clade: the specimens of Malawi and Tanzania cluster together in a group including *A. airensis*, *A. russatus* and *A. dimidiatus*, while the Transvaal specimen is basal to a non-Acomyinae clade (*Mus* and *Steatomys*) and the other *Acomys* species. Moreover, *A. wilsoni* is basal to all other *Acomys* specimens, except *A. subspinosus* which is mislocated in the outgroup. This tree reveals a high level of homoplasy.

CYTOCHROME B MOLECULAR PHYLOGENY

Cytochrome b sequences

The gene is 1141 bp long, coding for 380 amino acids (translation table: vertebrate mitochondrial genetic code), ending with an incomplete stop codon (single T completed in TAA by post-transcriptional polyadenylation). Some of the sequences are not complete: the two *A. wilsoni* (141 and 116 bp lacking at the beginning of the gene), the one from *A. subspinosus* (77 bp lacking after position 746), and the one from *Tatera kempi* which is not complete at its 3' extremity (1125 bp). In *Steatomys*, the gene is of same length as in *Acomys*, but has a complete stop codon (1143 bp, ending TAA).

Sequences from the two specimens from Berega (Tanzania) are almost identical (only one difference at pos. 471, A to G), and the two sequences from Mwenerondo (north Malawi) *Acomys* are completely identical. Thus, only one sequence from each of these localities was kept for the phylogenetic analysis. The nine new sequences considered for the analysis, plus the second sequence from Berega, are deposited in GenBank database (accession numbers AJ010554 to AJ010562, and AJ01024). Accession numbers of the 10 previously published sequences are: X96996, Z96051, Z96056, Z96058, Z96063 to Z96068.

Analysis of the phylogenetic molecular tree

The phylogenetic tree obtained with PAUP is shown in Figure 4E. The topologies of the trees obtained

under the three other algorithms are identical to the one presented in Figure 4E. In Puzzle, only one species can be chosen as outgroup at a time, but using the three changes rather than the topology and support values.

The phylogenetic tree is congruent with the previous study on cytochrome *b* (Barome *et al.*, 1998) which did not include *A. subspinosus* and *A. wilsoni*. In the tree presented here, *A. subspinosus* individualizes before *A. spinosissimus*, which would make it the most basal *Acomys* lineage, but the support for this branching is low (76-63/+10-73-51). However, such a basal position of *A. subspinosus* would agree with previous morphological analyses which describe the species as very plesiomorphic and quite different from other *Acomys* species. Denys *et al.* (1994) even proposed to assign it to a new subgenus *Subacomys*.

The sequences of *A. spinosissimus* from the four localities considered here (#2, 4, 7, 16) cluster together as a monophyletic group (support values: 95-99/+24-100-97). Within this *A. spinosissimus* clade, the sequence of *Acomys* from Berega (Tanzania) is the most basal, followed by the specimens from Mwenerondo (north Malawi), the remaining clade clustering the populations of Liwonde National Park (south Malawi) and Pafuri, Kruger National Park (South Africa). The sequencing of a fragment (400 bp) of the non-coding region of the mitochondrial DNA shows that the four *spinosissimus* specimens considered here share a 80 bp-long deletion, not present in any other sequence from eleven other *Acomys* species, including *subspinosus* and *wilsoni* (Barome *et al.*, 1998).

The *A. wilsoni* lineage splits after the *subspinosus* and *spinosissimus* lineages, but its relationship with *A. russatus* is not resolved, and the two species cluster in a trifurcation with the clade grouping the remaining species (*A. cahirinus*, *A. airensis*, *A. dimidiatus*, *Acomys* sp.).

CYTOGENETIC ANALYSIS

The male specimen from Berega (Tanzania) was karyotyped and its chromosomes were studied by different banding techniques (Fig. 5). The study revealed some remarkable karyotype features previously found by Matthey (1965) in *A. selousi*, namely a giant X (about 20% of the length of haploid chromosome set) and variation in diploid number between cells of the studied specimen (about one half had $2n=60$ and the other 61). By its banding pattern, the short arm of the X chromosome corresponds to the original type of X chromosome found in numerous species of *Acomys* (Volobouev, Tranier & Dutrillaux, 1991; Volobouev *et al.*, 1996a; Volobouev, Gautun & Tranier, 1996b), whereas the long arm is mostly C-band positive and late replicating. From C- and R-banding, there are

at least two other chromosomes partly or entirely heterochromatic, likely related to the sex chromosome system (Fig. 5). Here these chromosomes were always present, despite inter-cellular chromosome number variation within the individual.

DISCUSSION

TAXONOMY

A. spinosissimus was described from Mozambique localities Tete and Buio by Peters (1852), *A. selousi* from Bulawayo, Zimbabwe (De Winton, 1896), and *A. transvaalensis* from north Transvaal (Roberts, 1926). *A. subspinosus* was originally described by Waterhouse (1838) from the Cape of Good Hope (South Africa).

In his taxonomic revision, Ellerman (1941) considered *A. selousi* to be a separate species within the *cahirinus* group but did not allocate *A. spinosissimus* or *A. transvaalensis* to any group because he could not see the type specimens; however, he considered *A. subspinosus* to be a distinct species. Later, Ellerman, Morrison-Scott & Hayman (1953) grouped *spinosissimus*, *transvaalensis* and *selousi* as subspecies of *A. cahirinus*, but still recognized *A. subspinosus* as a distinct species. Matthey (1965) proposed *A. selousi* as a valid species on the basis of peculiar cytogenetic characteristics. In his revision of East African mammals, Kingdon (1974) included *selousi* as a race of *A. spinosissimus* and provided an identification key based on external characters. He described *A. spinosissimus* from Tanzania as having a dark brownish back with reddish sides of head and body. According to the same author, these specimens have a long nose and a nasal over 9.8 mm, and Setzer (1975) used this character as the key to identify this species. Petter (1983) confirmed the differences in the palate between *A. transvaalensis* and *A. selousi* mentioned by Roberts (1926). He observed a difference in the rigidity of the back bristles of *A. spinosissimus* and *A. selousi*, but noticed that these two forms could not be distinguished from dental morphology. In their revision, Dippenaar & Rautenbach (1986) considered *A. selousi* and *A. transvaalensis* to be synonyms of *A. spinosissimus*, and their morphometrical analysis revealed homogeneity of *A. spinosissimus sensu lato* specimens in populations from Mozambique to Transvaal.

Smithers & Wilson (1979) observed geographical variation in the dorsal pelage coloration of *Acomys* from Zimbabwe (reddish-brown, darker to paler from east to west), and they also mentioned that colour variability depended on the time of year in two localities; Dippenaar & Rautenbach (1986) observed "a clinal gradation of the amount of grey in dorsal pelage along a south-north axis". From biometric analysis, Dippenaar & Rautenbach (1986) also found a clinal variation along a south-north axis for the cranial

shape and length of tooth rows, but concluded that "the recognition of subspecies will simply reflect the arbitrary choice of body color as being of diagnostic value". Nevertheless, these authors examined neither Tanzanian and Malawian specimens, nor the dental patterns that could be useful in taxonomic attribution and key-characters in *Acomys* (Denys *et al.*, 1994; Petter, 1983). Finally, according to Musser & Carleton (1993), *A. transvaalensis* and *A. selousi* are synonyms of *A. spinosissimus* and the species extends into NE Tanzania, SE Zaire and Malawi.

PREVIOUS CYTOGENETIC ANALYSIS

The standard chromosome analysis carried out by Matthey (1965) on *Acomys* specimens from South Malawi and northern Tanzania, attributed to *A. selousi* by Francis Petter (in Matthey, 1965), revealed peculiarities making this species one of the cytogenetically oddest among mammals. First, both males and females possessed a sole X chromosome, unusually large and representing 19% of the total length of haploid chromosome set. Second, the diploid chromosome numbers varied from 58 to 62 among the 14 specimens studied (13 from southern Malawi and one from Ifikara, Tanzania), and chromosome variation was also observed among cells of the same individual (Matthey 1965). Dippenaar & Rautenbach (1986) carried out standard chromosome analysis of nine *A. spinosissimus* specimens from Transvaal. At first glance, their karyotype looks similar to that of *A. selousi* from Malawi-Tanzania, by having the same diploid number of 60, ten bi-armed chromosomes, many acrocentrics and a large X chromosome. A detailed comparison shows that there are a number of important differences between them. First, the chromosomal mechanism of sex determination in *A. spinosissimus* is of XX/XY type, whereas in *A. selousi* both males and females possess a sole X chromosome. Second, *A. spinosissimus* is characterized by a stable karyotype, i.e. diploid number remains the same in all animals studied as well as among the cells of the same individual, whereas *A. selousi* is characterized by inter- and intra-individual chromosome variation. Third, the X chromosome of *A. spinosissimus* represents 10% of the total length of haploid set and is thus about twice shorter than the X of *A. selousi*.

These important cytogenetic differences suggest that *spinosissimus* and *selousi* could be two distinct species despite the conclusions of Dippenaar & Rautenbach (1986).

COMPARISON BETWEEN MORPHOLOGICAL AND CYT B PHYLOGENIES

The analysis of morphology and cytochrome *b* sequences yield incongruent results. The morphological

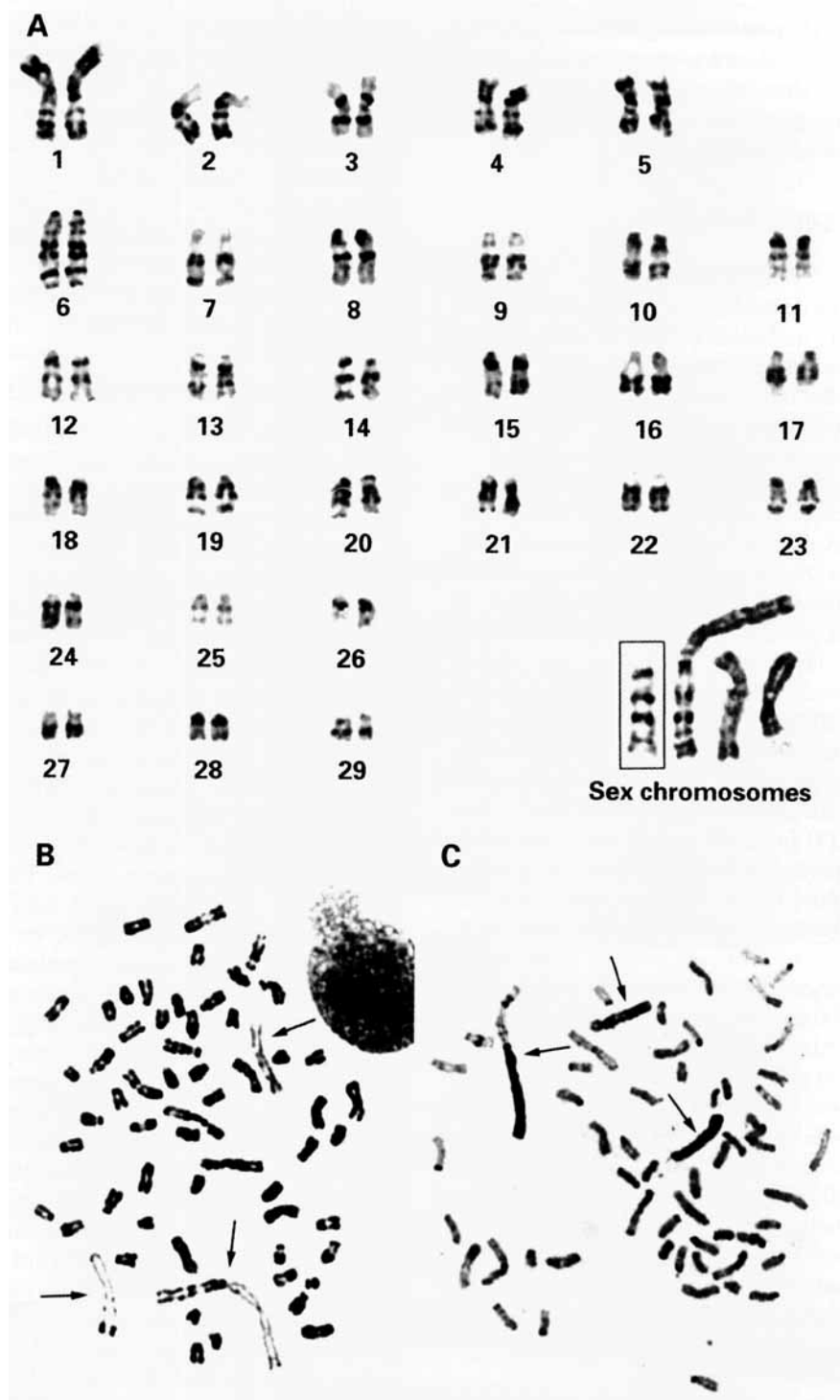


Figure 5. A, G-banding (GTG) karyotype of *A. cf. selousi* from Berega (1998-1611) with 61 chromosomes comprising 10 metacentrics, 48 acrocentrics and three sex chromosomes. In the insert the X chromosome of *A. cahirinus* is presented for comparison. B, R-banding (RBG) metaphase; C, C-banded (CBG) metaphase of *A. cf. selousi*. The 'sex' chromosomes are arrowed (see text for explanation).

analysis does not resolve the relationships between *A. subspinosus* and *A. wilsoni*; the *cyt b* phylogeny gives *A. subspinosus* as the most basal *Acomys* species,

followed by the monophyletic group *A. spinosissimus*, while *A. wilsoni* and other species are more apomorphic. However, one of the two morphological trees

and the *cyt b* phylogeny both argue in favour of the monophyly of *A. spinosissimus sensu lato*. The first morphological tree (Fig. 4C) confirms the close relationship between *A. subspinosus* and *A. spinosissimus* proposed in the Denys *et al.* (1994) scenario (Fig. 4B). The second morphological tree (Fig. 4D) suggests *A. wilsoni* as a sister taxon of group including *A. cahirinus*, *A. dimidiatus*, *A. ignitus* and *A. sp* like the *cyt b* tree (Fig. 4E). Both morphological and molecular trees agree in distinguishing *A. cahirinus* from *A. dimidiatus*.

On the basis of the first morphological tree (Fig. 4C), one could attribute specimens from North Malawi, Tanzania and Mozambique to *A. spinosissimus sensu stricto*, those from Zimbabwe to *A. selousi sensu stricto* (as sister group), and those from South Africa to a more basal *A. transvaalensis*. The first group would not be supported by *cyt b*, since the sequences from Tanzania and Malawi do not cluster together as a monophyletic group, the sequence from Transvaal specimen reveals that it is a sister group to that from Malawi. The second morphological tree illustrates the extensive morphological differences between the Transvaal *Acomys* from Pafuri and the species from Malawi, which could suggest that the Transvaal specimen belongs to a distinct species from the other *A. spinosissimus sensu lato*. This result is not supported by the *cyt b* phylogeny for two reasons: all *A. spinosissimus* cluster in a strongly supported monophyletic clade, and the Pafuri specimen is in apomorphic position within this clade.

Cytogenetics would argue in favour of the attribution of the Malawian and Tanzanian specimens to *A. selousi* and of the Transvaal specimens to another species (*A. spinosissimus* and/or *A. transvaalensis*), but this would not be supported by either *cyt b* phylogeny or morphology. The results presented here suggest that *A. spinosissimus* should be considered as a valid species, which includes the *selousi* and *transvaalensis* intra-specific forms. Additional cytogenetical and *cyt b* data about *A. selousi* from the Zimbabwe type locality as well as from Mozambique and other intermediate regions are now needed.

COMPARISON BETWEEN PHYLOGENY OF CYT B AND ANALYSIS OF PROTEIN POLYMORPHISM

The phylogenetic tree obtained with *cyt b* (Fig. 4E) can be compared with the phenogram based on protein polymorphism (Fig. 4A) from Janecek *et al.* (1991); several of the *Acomys* specimens in the present were also studied by Janecek *et al.* (1991): one of the *A. spinosissimus* (loc. 16), the *A. subspinosus* (loc. 21), the two *A. wilsoni* (locs. 23 and 24), the two *A. ignitus* (locs. 25 and 26), the two *A. dimidiatus* (locs. 27 and 28) and one *A. russatus* (loc. 29), respectively numbered

1, 2, 20, 21, 5, 7, 12, 14 and 11 in Janecek *et al.* (1991).

In Janecek *et al.* (1991), the species *A. spinosissimus* and *A. subspinosus* were considered as outgroups for a detailed sampling of other *Acomys* species (mostly *A. wilsoni* and *A. ignitus*). The phenogram obtained by these authors shows that the two species from southern Africa are very divergent from the other species, but the tree was unrooted. It shows that *A. ignitus* and *A. wilsoni*, from the same localities cluster in two distinct groups, *A. wilsoni* being sister group to *A. russatus* from Jordan, and *A. ignitus* closely related to *A. cahirinus* from the Levant. The specimens called *A. cahirinus* by Janecek *et al.* (1991), should in fact be called *A. dimidiatus*, arguments about this coming from cytogenetics (Volobouev *et al.*, 1996a,b) and *cyt b* phylogeny (Barome *et al.*, 1998; Barome, Monnerot & Gautun, 2000). Thus the phenogram of Janecek *et al.* (1991) is congruent with the cytochrome *b* tree presented in Figure 4E, except that the relationship between *A. russatus* and *A. wilsoni* is not resolved in this tree.

DIVERGENCE TIMES

Because of the poor *Acomys* fossil record, calibration of a molecular clock for this genus is not possible, so the divergence rate for *cyt b* between *Mus* and *Rattus* was used to calibrate that of *Acomys*. This method has two sources of error. First, the date of splitting of the lineages *Mus* and *Rattus* is itself an estimate: between 8 and 12 Myr in Jaeger, Tong & Denys (1986), or 12 and 14 Myr (see discussion in Robinson *et al.*, 1997), older dates have also been proposed (O'Uigin & Li, 1992; Kumar & Hedges, 1998). Second, the rate of evolution of a given gene may be different in two lineages; however, the equivalence of *cyt b* evolutionary rate was tested (Tajima, 1993) for different *Acomys* lineages versus *Mus* and versus *Rattus*, and was not found to be significantly different (Barome *et al.*, in press). We thus chose to use the interval of 10–14 Myr for the *Mus*–*Rattus* split.

Two *cyt b* sequences were considered from *M. musculus* (Bibb *et al.*, 1981; Shan, Vazquez & Lewis, 1990), one from *Nannomys musculoides* Temminck 1853 (Barome *et al.*, 1998) which belongs to the *Mus* lineage (Catzefflis & Denys, 1992; Barome *et al.*, 1998), and three from *Rattus norvegicus* Berkenhout 1769 (Goertz & Feldmann, 1982; Koike *et al.*, 1982; Gadaleta *et al.*, 1989); no complete *cyt b* sequences from other taxa in the *Mus* or *Rattus* lineage were available. Divergences between the two lineages were thus the mean of *Mus*–*Rattus* and *Nannomys*–*Rattus* divergences (mean 19.94%; divergence values used are Kimura-2 parameter distances, Kimura, 1980). Using the interval 10–14 Myr, the divergence rate is thus between 1.38 and 1.93%.

Table 2. Date estimates for each cladogenesis event that occurred between *Acomys* clades in the cyt *b* phylogenetic tree (Fig. 4E)

Cladogenesis event considered (nodes in trees Fig. 4D and 4E except for <i>Mus–Rattus</i> lineages)	Mean nucleotide divergence ^a	Estimated date of divergence in Myr ^b	
		<i>Mus/Rattus</i> split at 10 Myr	<i>Mus/Rattus</i> split at 14 Myr
<i>Mus/Rattus</i> lineages	0.199	10.0	14.0
<i>subspinosus</i> /sister clade (16) ^c	0.222	11.1	15.6
<i>spinosissimus</i> (4)/sister clade (12)	0.213	10.7	15.0
Tanzania/other <i>spinosissimus</i> (3)	0.152	7.6	10.7
N. Malawi/[Transvaal + S. Malawi]	0.129	6.5	9.1
Transvaal/S. Malawi	0.117	5.9	8.2
Trifurcation [<i>wilsoni</i> (2)]/[<i>russatus</i> (2)]/sister clade (8)]	0.186	9.3	13.0
<i>ignitus</i> (2)/sister clade (6)	0.155	7.7	10.9
[<i>cahirinus</i> + <i>airensis</i>]/[<i>dimidiatus</i> (2)-sp. (2)]	0.103	5.2	7.2
<i>dimidiatus</i> (2)-sp. (2)	0.093	4.7	6.5

^a Kimura-2 parameter distance (Kimura, 1980).^b Ratio of mean nucleotide divergence between considered sequences by mean *Mus–Rattus* lineages divergence for cyt *b*, multiplied by 10 or 14 (see text).^c Numeral in brackets is the number of sequences in the group considered.

Mean divergence between the *A. subspinosus* sequence and each of the 16 other *Acomys* sequences is 22.20%, so the estimated date for splitting of the *A. subspinosus* lineage is about 11.1–15.6 Myr. To estimate the date of separation between the *A. spinosissimus* and its sister clade, the mean of all taxa distances was calculated between them. The value obtained is 21.31%, which gives a date of 10.7–15.0 Myr. Following the same method, divergence times were estimated for all branching nodes of the phylogenetic tree and are listed in Table 2. Kimura-2 parameter distances have been calculated including all kinds of substitutions.

The date of emergence of the *A. subspinosus* lineage is close to previously proposed dates of origin for *Acomys*: using DNA/DNA hybridization, Catzefflis *et al.* (1993) found the time of divergence for *Acomys* from other Murinae and Gerbillinae to be about 13–15 Myr. On the basis of these results plus palaeontological data, Denys *et al.* (1994) then proposed a date between 7 and 14 Myr.

The divergence between *A. subspinosus* and the other *Acomys* species is of the same order as the *Mus–Rattus* divergence, which means that either the evolutionary rate in *Acomys* cyt *b* is higher than the one in *Mus* and/or *Rattus*, or the origin of the genus *Acomys* is indeed more ancient than the split between the two *Mus* and *Rattus*. As the rates were found to be similar (see above), this second hypothesis is more likely.

BIOGEOGRAPHIC HYPOTHESIS FOR *ACOMYS* DISPERSION IN SOUTHERN AFRICA

Two biogeographical hypotheses for the colonization of southern Africa by *Acomys* can be formulated from the conclusions presented above.

In the first hypothesis, the centre of origin of *Acomys* is the southern part of the African continent. From this ancestral stock, *A. subspinosus* would have differentiated while the other lineage was migrating northwards and giving birth to *A. spinosissimus*, *A. wilsoni* and other species. The distribution of *A. subspinosus*, nowadays restricted to the Cape region, seems to argue in favour of this hypothesis. It has been shown that two other murine genera have been extending their distribution area from the Transvaal to Tanzania during Plio–Pleistocene times. *Otomys* and *Aethomys* dispersed northwards between 2.8 and 1.7 Myr (Denys, 1999), and dispersion of *Acomys* could have occurred slightly earlier along the same route. This scenario fits with the fossil record which shows the presence of *A. mabele* close to *A. spinosissimus* around 5 Myr in the Cape region (Denys, 1990). Later, the extinct species *A. coppensi* is recorded around 3.3–3.5 Myr in Hadar, Ethiopia (Denys *et al.*, 1994), and *A. spinosissimus* appears around 3 Myr in caves in Transvaal (Denys, 1999). However, the *Acomys* fossil record is very sparse, and more work is needed.

In the second hypothesis, the ancestral stock of

Acomys is located in East Africa (Kenya and Ethiopia) as proposed by Matthey (1968). From here the genus would have spread along two paths, one to the Mediterranean region and the other to the south of the continent, giving birth to *subspinosus* and subsequently *spinosissimus*, and splitting into individual lineages from north to south (Tanzania, northern Malawi, southern Malawi, South Africa). This is supported by the branching of populations within the *A. spinosissimus* clade in the cytochrome *b* phylogeny, but it implies the poorly parsimonious postulate that *A. subspinosus* is the result of a first north-south wave of colonization, prior to the *A. spinosissimus* expansion.

The speciation process could also have been caused by a progressive fragmentation of the distribution area, resulting from the formation of the Rift Valley in its south-western part (Denys, 1999), or from the extension of tropical forests zones dividing the savannah domain into mosaic blocks during climatic fluctuations and isolating the different species by vicariance.

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APPENDIX 1

Skull measurements of the *A. spinosissimus* specimens used in this study.

Tanzania	HB ^a	TL ^b	HF ^c	E ^d	LGT ^e	WZYG ^f	LNAS ^g	WNAS ^h	CIO ⁱ	LFPAL ^j	LBT ^k	LS13 ^l	LI13 ^m	LMDB ⁿ	HMDB ^o
nhm 761 559					25.65	12.53	10.34	3.2	4.73	5.46	4.61	4.2		17.2	6.47
761 560					26.57	12.67	10.42	2.84	4.99		4.42	4.42	4.3	17.03	
761 561					26.83		10.6	3.17	4.88	5.72		4.46	4.01	16.53	6.57
761 562					26.66	12.68	11.03	2.84	4.74		4.48	4.13	4.04		
761 563					27.97	13.18	11.03	2.82	5.02	5.31	4.65	4.58	4.3	16.92	
Berega mnhn	91	73	15.1	14.7	25.22	12.24	9.39	2.1	4.55	5.55	4.24	4.12	3.7	15.86	5.56
1998-1611															
mnhn 1964-53	80		17	15	24.35	11.98	9.56	2.28	4.9	5.19	4.88	4.21	4.06	15.38	
nhm 84.1842					26.37	12.85	10.42	2.81	4.74	5.23	4.05	4.31	4.2	16.94	6.28
Mean					26.33	12.59	10.35	2.76	4.82	5.41	4.48	4.30	4.09	16.54	6.44
Malawi	HB	TL	HF	E	LGT	WZYG	LNAS	WNAS	CIO	LFPAL	LBT	LS13	LI13	LMDB	HMDB
Mwener. 96 36	94.5	87	16.5 + 1.5	16											
Mwener. 99-74	84.5	89	16 + 1	14	25.38	11.62	10.33	2.96	4.49	5.63	3.96	4.29	3.74	16	
nhm 97 10 1					25.7		10.51	2.64	4.64	5.91	4.2	4.33	3.92		
208															
97 10 1 285	85	83	16.5		25.47	13.08	9.53	2.95	4.6	5.47	4.56	4.24	4.06	16.4	6.07
97 10 1 286					25.4	12.8	9.31	2.73	4.94	5.37	4.08	4.38	4	16.42	5.89
97 10 1 287	86	77	16		25.13	12.31	9.74	2.7	4.48	5.47	4.48	4.03	3.94	16.19	6.32
nhm 49 378					25.8	13.12	9.48	2.38	4.46	5.7	4.6	4.05	3.64	16.69	5.7
TM 41760							9.61	2.86	4.62	5.41		3.92	3.66	15.95	6.05
TM 41761					24.18	11.29	8.41	2.63	4.72	5.51	4.62	4.12	3.72	15.06	5.18
TM 41762					12.76		9.95	2.84	4.77	5.44		4.13	3.65	15.35	5.8
mnhn 1964-303	90		18	14	25.08				4.84	6.03	4.74	4.13	3.58	16.11	
1964-304	95	73	18	14	26.41		11.43	3.21	4.85	6.27	4.87	4.18	3.95	17.09	7.28
1964-883	90		18.5	15	27.68	13.45	11.56	2.83	4.74	6.39	4.67	4.38	4.03	17.59	7.28
1964-884	90		18	14	27.4	12.85	10.79	3.02	4.87	6.47	4.68	4.47	3.94	16.96	6.24
1964-885	90		17.5	14	27.1	13.61	10.27	3	4.85	6.04	3.9	4.45	4.05	17.67	6.18
1964-886	90		18	14	26.06	13.39	10.56	3.17	4.83	5.73	4.8	4.21	4.04	17.28	5.88
1965-887	90	85	18	13.5	27.02	12.51	10.49	2.69	4.83	6.36	4.76	4.2	4.07	17.26	6.03
1964-888	85		18	15	26.97	12.59	9.81	2.75	4.75	5.94	4.94	4.18	4	17.22	5.86
1964-889	80	65	18	13.5	26.35	12.05	9.41	2.56	4.64	5.61	4.25	4.24	3.96	15.89	5.88
1964-890	80	73	17.5	14	25.93		9.74	2.71	4.97	5.55	4.42	4.24	3.85	16.35	6.28
1964-891	80		17	12.5	23.96	10.73		2.47	4.63	5.39	4.85	4.37	4.07	15.08	
nhm 14 6 13 31	99	83	19.5	15	27.48		10.83	2.86	5.19	6.49	4.55	4.53	4.16	17.83	6.42
14 6 13 32	98	70	19	14	25.4		9.73	2.28		5.9	4.87	4.23	4.02	16.9	5.91
Mean	88.6	78.5	17.83	14.1	26.00	12.54	10.07	2.79	4.75	5.82	4.54	4.24	3.91	16.54	6.06

continued

APPENDIX 1 – continued

Zimbabwe	HB	TL	HF	E	LGT	WZYG	LNAS	WNAS	CIO	LFPAL	LBT	LS13	LI13	LMDB	HMDB
mnhn 1990 592	82	82			25.55		9.83	3.17	4.7	5.85	4.64	4.07	3.62	16.11	5.8
mnhn 1990 352	72	80			24.35		8.79	2.62	4.41	5.35	4.07	3.76	3.62	15.46	5.66
97 1 4 44 type	83	92.5	16.5	14	25.5	12.5	10.5	3	4.6	5.5		4	3.6	13.1	
nhm 97 1 4 45	70	79	17		24.98	12.36	9.6	2.72	4.83	5.17	4.88	3.01	3.58	15.59	6.03
97 1 4 46	70	79	18		24.18		8.76	2.69	4.64	5.43	4.38	3.87		15.17	5.49
97 1 4 47					24.4	11.89	8.68	2.63	4.69	5.29	4.51	3.96	3.42	15.53	5.67
97 1 4 48					26.61		10.39	2.83	5.01	5.94	4.83	4.34	3.82	17.31	6.53
Mean	75.4	82.5	17.17	14	25.08	12.25	9.51	2.81	4.70	5.50	4.55	3.86	3.61	15.47	5.86
Mozambique	HB	TL	HF	E	LGT	WZYG	LNAS	WNAS	CIO	LFPAL	LBT	LS13	LI13	LMDB	HMDB
ZMB 1711 type	100	85	18	13.5	27.1		10		4.5			3.85	3.7	13.4	
nhm 7 6 2 80	96		17	13	25.52	11.89	9.2	2.88	4.53	5.04	4.91	4.15	3.61	15.65	5.9
7 6 2 81	93	87	17	15	25.27	12.29	9.21	2.86	4.44	5.72	4.54	4.22	3.89	15.79	6.04
7 6 2 82	91	78	16	12	25.59	12.52	9.23	2.73	4.77	5.61	4.14	3.75	3.62	15.9	5.84
7 6 3 86	97	67	19	13.5	25.97	12.97	9.67	3.06	4.72	5.3	4.26	4.12	3.78	16.46	5.92
7 6 2 87					25	12.15	9.12	2.69	4.52	5.54	4.55	4.01	3.64	15.42	5.92
7 6 3 89	82	69	17		23.44	10.6	8.31	2.6	4.35	5.17	4.04	4.01	3.83	14.57	5.23
Mean	93.1	77.2	17.33	13.4	25.41	12.07	9.25	2.80	4.55	5.40	4.41	4.02	3.72	15.31	5.81
South Africa	HB	TL	HF	E	LGT	WZYG	LNAS	WNAS	CIO	LFPAL	LBT	LS13	LI13	LMDB	HMDB
TM 29879	93		17	13	24.52	11.59	9.84	2.74	4.67	5.38	3.9	4.01	3.6	15.22	
TM 30520	94	86	17	14	24.15	12.34	9.71	3.16	4.65	6.07	3.94	3.93	3.4	15.19	5.86
TM 30577	86	68	16	14	23.88	11.28	9.27	2.99	4.45	4.17	3.81	3.83	3.4	15.16	5.16
TM 30578	95		17	18	23.18	12.75	9.84	2.78	4.65	5.8	3.46	3.96	3.38	15.8	5.55
TM 30579	86	76	16	14	24.08	11.72	9.3	2.61	4.4	5.01	4.08	4.04	3.71	15.11	
TM 30614	90		18	13	22.72	10.72			4.46	4.79	4.47	3.94	3.61	14.13	5.51
TM 30615	88		15	14	23.85	11.42	9.66	2.94	4.37	4.98	3.71	3.94	3.79	14.59	5.44
TM 30616	96	80	15	16	25.08		9.32	2.9	4.67	5.12	3.78	3.75	3.34	15.12	6.18
TM 3593 type	85		15	12.5	25	13	9.6		4.6	5.3		3.8	3.4		
Mean	90.3	77.5	16.22	14.28	24.27	11.85	9.57	2.87	4.55	5.18	3.89	3.91	3.51	15.04	5.62

*Head and body length; ¹tail length; ²ear length; ³total length of the skull; ⁴width of the skull at the level of zygomatic arches; ⁵nasal length; ⁶nasal width; ⁷inter-orbital constriction; ⁸length of the incisor palatal foramen; ⁹length of the tympanic bullae (maximum); ¹⁰upper molar row length; ¹¹lower molar row; ¹²length of the mandible; ¹³height of the mandible.

APPENDIX 2

Character states for the *A. spinosissimus* specimens examined here plus other *Acomys* specimens and respective outgroups (coding given in the text). * outgroups for second morphological tree (Fig. 4D), + outgroup of the first tree (Fig. 4C). In bold, ingroup taxa used for the first morphological matrix.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Malawi	1	2	1	1	1	1	0	2	2	0	0	0	0	0	0	1	1	1
Tanzania	1	2	1	1	1	1	0	2	2	0	0	1	1	1	0	1	1	0
Zimbabwe	1	1	0	1	1	1	0	2	1	1	0	0	1	1	1	1	1	0
Mozambique	1	1	1	1	1	0	0	2	2	0	0	0	1	1	1	1	1	0
Transvaal	1	1	0	1	1	0	0	2	1	1	1	1	0	0	0	1	1	0
<i>A. cahirinus</i>	1	2	1	1	0	0	0	2	2	0	0	0	1	0	0	0	0	1
<i>A. dimidiatus</i>	1	2	1	1	1	0	1	2	2	0	0	1	1	0	0	0	1	1
<i>Acomys</i> sp.	1	2	1	1	0	0	0	2	1	0	1	0	1	0	0	1	0	1
<i>A. airensis</i>	0	2	1	1	1	0	0	2	2	0	0	1	1	0	0	0	1	1
<i>A. russatus</i>	0	2	1	1	1	0	0	2	1	0	0	1	1	1	1	0	0	1
<i>A. ignitus</i>	0	2	0	1	0	1	1	2	2	0	1	0	0	0	0	0	1	1
<i>A. wilsoni</i>	0	0	0	1	0	0	0	1	1	0	0	0	1	1	0	0	1	1
<i>A. subspinosus</i>	1	1	0	0	0	0	1	0	0	0	1	1	1	1	1	1	0	0
<i>Uranomys</i> +	0	1	1	1	1	0	0	1	0	0	1	0	1	1	1	0	0	0
<i>Steatomys</i> * +	0	0	0	1	0	0	0	0	0	1	1	0	1	0	0	0	0	0
<i>Aethomys</i> +	1	2	1	1	1	0	0	1	0	0	1	1	1	1	1	0	0	1
<i>M. musculus</i> *	1	0	0	1	0	0	0	1	1	1	1	1	1	0	0	0	1	1
<i>Tatera kempi</i> *	0	2	1	0	0	0	0	0	0	0	1	1	1	1	1	2	2	0